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HEMATOPOIESIS AND BLOOD GROUPS

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Hematopoiesis

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Timelines :

	Age	Sites of hematopoiesis
Intrauterine life	3 weeks – 3 months	Yolk sac
	3 – 5 months (Hepatic stage)	Liver/spleen (physiological)
	5 – 9 months	Red bone marrow : Long & flat bones
	Birth – 20 years	
	> 20 years	Red bone marrow : Flat bones (midline) Eg : Iliac crest

ERYTHROPOIESIS :

Stages of erythropoiesis :

Pluripotent hematopoietic stem cell (PHSC)

↓ GATA, friend of GATA (FOG) } Transcription factors

Burst forming unit erythrocyte (BFU-E) : Earliest committed progenitor

↓

Colony forming unit erythrocyte (CFU-E) : Highest erythropoietin receptors

↓

Proerythroblast : Hb synthesis begins

↓

Basophilic erythroblast

↓

Polychromatophilic erythroblast : Hb appears

↓

Orthochromatic erythroblast : Hb synthesis complete

↓

Reticulocyte : Immature RBC

↓

Erythrocyte : Mature RBC (Size ~ 7.2 μ)

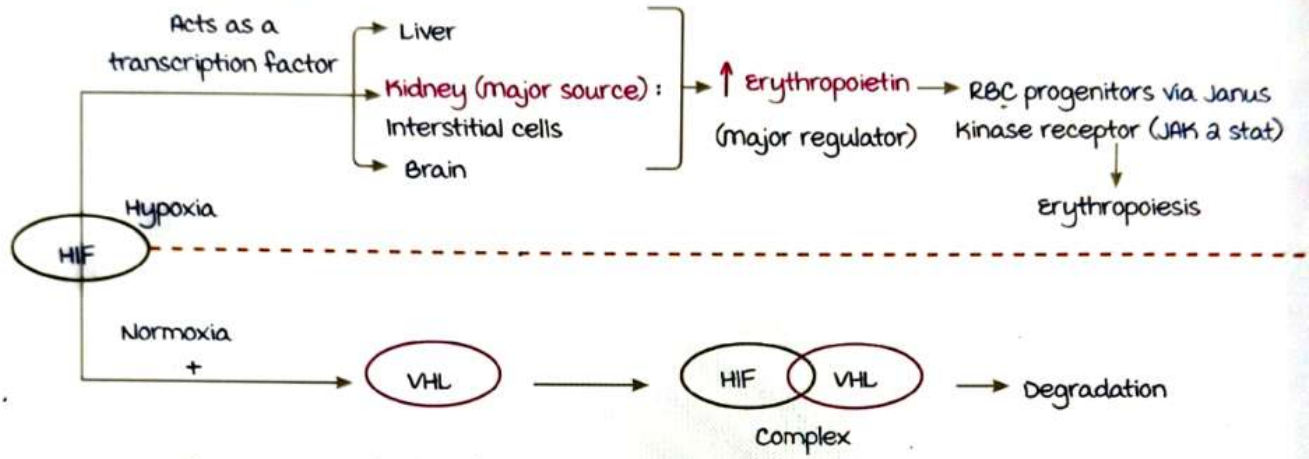
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Regulation of erythropoiesis :

Mechanism :



HIF : Hypoxia inducible factor.

VHL : Von Hippel-Lindau.

Factors regulating erythropoiesis :

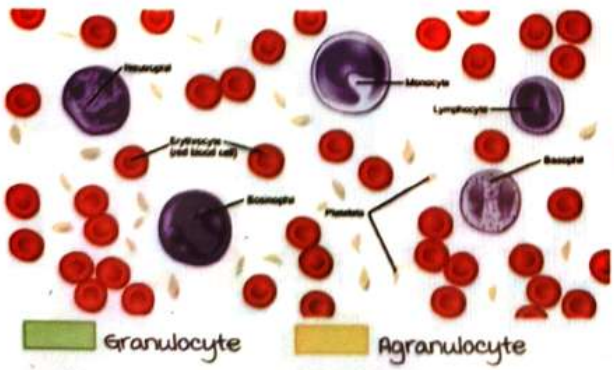
Factors

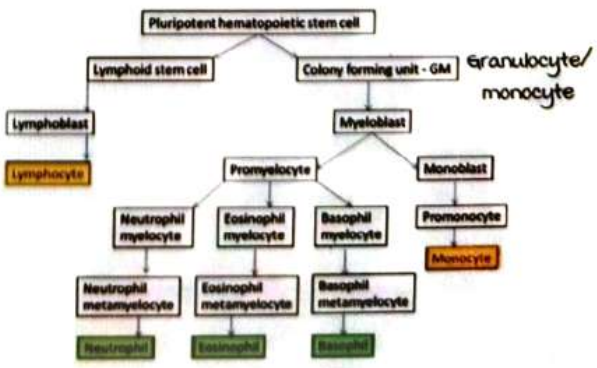
Promoting	Inhibiting
1. EPO.	Estrogen
2. Growth hormone	(Female : ↓ RBC count).
3. Cortisol.	
4. Thyroid hormone (T3).	
5. Testosterone.	

Clinical significance of EPO :

- Polycythemia vera :
  - Gain of function mutation : JAK a receptor → ↑ EPO → ↑↑RBCs.
  - Rx : Ruxolitinib (JAK a inhibitors).
- Drug forms : Epoetin alfa, darbepoetin alfa.
  - Indications : 1. Anemia of chronic disease (Eg : CKD).
    2. Dialysis.
- Blood doping : Misused by athletes to ↑ RBC count.

LEUCOPOIESIS :





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Regulation of leucopoiesis :

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Factors stimulating :

1. Colony stimulating factor (CSF) : Filgrastim.
2. Granulocyte monocyte colony stimulating factor (GM-CSF) : Sargramostim. } Rx of neutropenia
3. Granulocyte colony stimulating factor (GCSF).
4. monocyte colony stimulating factor (MCSF).
5. IL 5 : Eosinophils.
6. IL 3, 4 : Basophils.
7. Tumor necrosis factor (TNF).

THROMBOPOIESIS :

PHSC → GM-CSF thrombocyte → megakaryocyte → Platelets

Regulation :

Factors stimulating	Analogues
Thrombopoietin	Eltrombopag Romiplastin
IL-11	Oprelvekin

Blood Groups.

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ABO BLOOD GROUP SYSTEM :

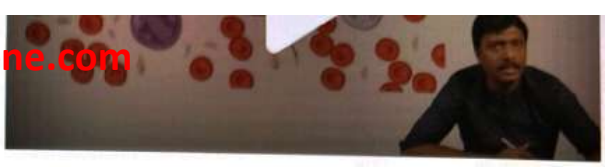
- Based on antigens & oligosaccharides in RBC.
- Antigens : Agglutinogens.
- Antibodies : Agglutinins.

Types :

Features	A	B	AB (Universal recipient	O (Universal donor)	Bombay
Sugars	NAG	GAL	Co dominance : Both antigens are expressed		
Antigen/ Agglutinogen	A, H	B, H	A, B, H	H	
RBC	RBC	RBC	RBC	RBC	RBC
Antibody	anti B	anti A	-	anti A anti B	-

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- Antigens : H, A, B
- Sugars : N-acetyl galactosamine (NAG).
  Galactose (GAL).
- Landsteiner's law : If an antigen is present in RBC its corresponding antibody must be absent from serum.

Blood group antigen sites :

1. RBC.
2. Saliva.
3. Amniotic fluid.
4. Semen.
5. Intestinal bacteria.
6. Food.

RH SYSTEM :

Antigens : C, D, E.

Types :

Based on presence of D antigen

- Rh+ : D antigen +
- Rh− : D antigen −

Clinical significance :

If mother is Rh − & baby is Rh +
↓
Fetal blood enters maternal circulation
↓
Stimulates production of maternal antibodies against fetal RBC
↓
Hemolysis in subsequent pregnancy
↓
Erythroblastosis fetalis/ Rh incompatibility
↓
↑ bilirubin
↓
Unconjugated bilirubin : Crosses blood brain barrier
↓
Deposited in basal ganglia
↓
Kernicterus

Rx : Anti Rh antibodies.

OTHER BLOOD GROUP SYSTEMS :

Kell system	If absent → Mcleod phenotypes : Acanthocytes.  Cardiac defects.
P system	• Autoantibodies against P → Paroxysmal cold hemoglobinuria. • Receptor for entry of : Parvovirus B19, E coli.
Duffy system	Receptor for entry of plasmodium vivax → Malaria

Feedback